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In this age of upgrades, let's say you have just acquired a 250+ GB hard disk and your poor old 20, 40 or 80 GB hard disk is packed and put into your cabinet. It is time now to dig it out and convert it into an external hard drive with minimal investment. You no longer need to convince your friend to open up his cabinet so you can hook up your hard disk to this machine to copy songs—just plug it into the USB port and it will get recognised as a mass storage device!

What you need

- ▣ Your old (a new one is also fine!) hard disk.
- ▣ An IDE to USB converter.

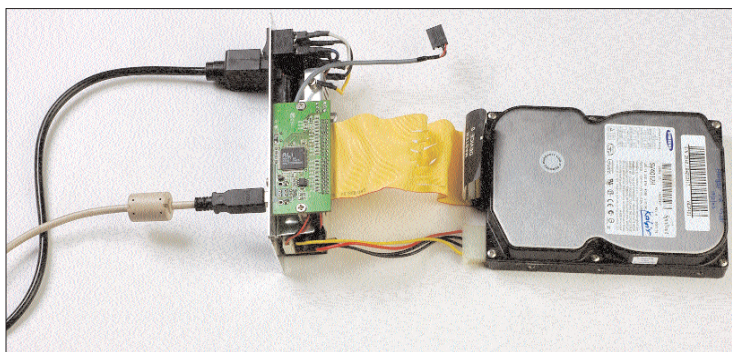
IDE to USB converters are available for the standard 3.5-inch desktop internal drive as well as the 2.5-inch laptop drives. When you are shopping for the converter, be sure to ask for a 3.5-inch form factor converter. The converter has an IDE and molex power cable that connects to the hard disk and a USB cable to connect to the USB port. There is an internal circuitry that converts the IDE interface to the USB interface. The power supply for the hard disk is drawn from a power adaptor bundled with the kit

Connect the IDE and power cable of the converter to the hard disk and secure the hard disk on the chassis by mounting it (if it is a case type, rather than open type). Plug in the USB cable to the USB port (USB 2.0 if you want the best read / write speed). Under Windows 2000 or Windows XP, the drive will be recognised as a mass storage device, just like a thumb drive. If in case the drive is not detected, unplug and plug it again. If prompted to, restart the PC.

To prevent chances of data loss, do not disconnect the hard disk while data is being written to it. For greater safety, 'stop' the device from the system tray before unplugging it.

There is more to an IDE to USB converter than making an external hard disk. If your laptop's CD ROM Drive is not working and you need to install a software from a CD urgently, you can plug in a CD-ROM drive just like a hard disk! Windows may need to restart after seeing a 'non hard disk USB device'. The best part of this setup is the price—an IDE to USB converter with casing will cost you just about Rs 500. A worthy investment for the functionality it provides!

If your PC has a USB 1.1 only interface, you can still connect your hard disk to it, but the read / write speed will be very slow.



The insides of an IDE to USB converter

Hard Disk Maintenance

We have learnt in our Physics classes that anything that moves, causes friction and hence wear and tear. The hard disk is one component that is in constant motion, with its spindle spinning at 7,200 rpm! This is why we keep talking about hard disk maintenance and cooling—here we will summarise the most important steps that you can take to ensure that the *old faithful*—your hard disk serves you for a long time!

Disk Defragmenter And Error Checking

The most basic of disk maintenance tasks still hold good. Just as a messy room makes it difficult for you to find things when want, the hard disk has to do more work to fetch data if it is scattered all across the disk. This is why you need to keep your disk defragmented. Go to **Start > Programs > Accessories > System Tools > Disk Defragmenter**; or right click on the partition (or drive) you wish to defragment and click on **Properties** and then **Tools > Defragment**.

Scandisk performs a write test on every cluster and reports an error if there is any. One must remember that disk errors cannot be cured by scandisk, but only identified and marked such that the affected areas are not used to store files. If scandisk shows bad sectors, its time to buy a new drive!

Windows XP calls the Scandisk operation 'Error Check-

Bet You Didn't Know

Centre An Object

If your BIOS supports booting from USB, you can even run an operating system from your just assembled external hard disk.

Samsung Hard Disk Drives with SATA

Satisfy your need for speed.













